

Comparing ICD Codes and Traditional Natural Language Processing to Identify Acute Pediatric Firearm Injury

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Objectives: Firearm injuries are a leading cause of pediatric mortality, but may not be adequately identified by International Classification of Diseases (ICD) codes. Natural language processing (NLP) could improve identification. We derived an NLP model from medical narratives and compared the sensitivity and specificity of ICD codes and the NLP models to identify pediatric acute firearm injury (AFI).

Methods: Retrospective study of ED visits by patients (aged 0 to 17 y) presenting to 7 pediatric emergency departments (ED) with a potential AFI between 2011 and 2019. A medical narrative review was performed to identify AFI narratives. These narratives, plus randomly selected narratives, underwent language feature extraction with logistic regression to identify topics significantly associated with the categories. Two Random Forest models were created: model 1 on all narratives associated with a potential AFI and model 2 using the confirmed AFI narratives. Model 2 was then applied to a new, unreviewed selection of random narratives. A second round of medical narrative review was performed on the random narratives, including those identified by model 2 as AFI. Sensitivity and specificity of ICD and NLP for the identification of AFI were calculated.

Results: A total of 2407 ED visits for potential AFI were identified.

Of those, 1183 narratives were annotated by narrative review as AFI. In the randomly selected narratives (n=1150), 47 additional AFI were identified, 16 by NLP and 31 by narrative review, for a total of 1230 AFI. ICD codes correctly identified 1171 (95.2%) AFI, with a sensitivity of 95.2% (93.8% to 96.3%) and specificity of 93.5% (92.6% to 94.4%). NLP correctly identified 1041 (84.6%) AFIs in the final data set with a sensitivity of 84.6% (82.5% to 86.6%) and specificity of 94.7% (93.8% to 95.5%).

Conclusions: ICD and NLP both had acceptable sensitivity and specificity for identifying pediatric AFI. Future directions include evaluating newer NLP models to address false positives and the identification of the intent of injury.

Key Words: firearm, pediatric firearm injury, natural language processing, International Classification of Diseases codes

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Firearm injuries are the leading cause of pediatric death in the United States.¹ Multiple professional organizations have called for increased attention to the public health crisis of firearm injury in America.^{2,3} This call urges expanded research in firearm injury and prevention, especially considering the lack of publications and limited research funding dedicated to this issue compared with other leading causes of mortality.⁴

When conducting research using clinical data, many researchers use the International Classification of Diseases (ICD) codes to identify disease states. ICD codes are vital for research using large data sets where other methods of identifying patients would be impractical.⁵ However, there have been reports challenging the accuracy of ICD codes for identifying disease states.⁶ The accuracy of ICD codes for identifying firearm injury is less defined, with studies focusing on the accuracy of ICD-documented intent rather than the injury itself.^{7,8} An accurate and facile method of identifying firearm injuries would be useful for research and advocacy purposes.

Previously, the only alternative to using ICD codes was by manual chart review, a process requiring substantial time and money.⁹ Natural language processing (NLP), however, offers the ability to use free text data in the medical record without the time and cost of manual chart review.^{10,11} NLP has been used for research in some medical disciplines, including pediatric injury, but has not been specifically evaluated in acute pediatric firearm injury.^{12–14} Therefore, the purpose of this study was to derive an NLP model from electronic health record narratives and compare the sensitivity and specificity of ICD codes and NLP models to identify acute pediatric firearm injury.

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METHODS

This was a retrospective study of ED visits by patients (aged 0 to 17 y) presenting to 1 of 7 Pediatric Emergency Care Applied Research Network (PECARN) emergency departments (EDs) with a potential firearm injury between January 1, 2011, and December 31, 2019, included in the PECARN Registry.¹⁵ Discrete elements and narrative data from the PECARN Registry included 4 sites from 2011 to 2019 (Children's Hospital of Philadelphia, Children's National Hospital, Cincinnati Children's Hospital Medical Center, Children's Hospital Colorado) and 3 additional sites from 2016 to 2019 (Ann & Robert H. Lurie Children's Hospital of Chicago, Children's Wisconsin, Nationwide Children's Hospital) in diverse regions across the United States. The study was approved by the institutional review boards of all involved sites.

ED visits were identified from the PECARN Registry and grouped into 2 data sets (Fig. 1) to derive the NLP models and assess test characteristics:

1. ICD data set: this data set contained both deidentified potential firearm injury ED visit narratives and general ED visit narratives. Potential firearm injury narratives were identified from the larger PECARN Registry database by ICD-9 or ICD-10 codes for firearm injury (Supplemental Table 1, Supplemental Digital Content 1, <https://links.lww.com/PEC/B698>).^{16,17} To identify most potential firearm visits, ICD codes for "other and unspecified firearms" (eg, BB guns, air guns, paintball guns) and subsequent visits after a primary firearm injury were included (Supplemental Table 1, Supplemental Digital Content 1, <https://links.lww.com/PEC/B698>). The general ED visit narratives were a randomly selected subset of 2469 from over 4 million remaining narratives in the PECARN Registry data set. These narratives could be associated with any diagnosis, including firearm injuries, without an associated ICD code.

2. Annotated data set: this data set was constructed from the potential firearm injury narratives from the ICD data set by structured review of ED narratives by the authors (K.D., M.F., M.A.), using a standardized template and training to determine whether an acute firearm injury occurred. Narratives were reviewed by 1 author unless the reviewer was undecided whether a firearm injury occurred. Those narratives were reviewed by the remaining authors to reach consensus. Visit narratives were defined as acute only if this was the first ED visit for firearm injury, and that visit occurred within 1 week of the firearm injury. We included transfers of care from other hospitals for acute injuries. Visit narratives for nonpowder weapons (eg, air guns, BB guns, paintball guns, Nerf guns) were marked as not an acute firearm injury.

Language Feature Extraction

All electronic health record medical narratives derived from the ICD data set were computationally split into tokens (words, numbers, etc.) using HappierFunTokenizer, version 1.5.7, available with the Differential Language Analysis Tool Kit (DLATK), an open-source Python package.¹⁸ Each medical narrative was represented as a set of language features. Two authors (S.J. and S.C.G.) labeled the top 100 most frequent words in the review text as stopwords ("the," "a," "an," etc.) and removed them from further analyses. They then extracted 2 features: a bag-of-words representing word multiplicity and topics to represent each narrative to analyze significant differences between acute firearm injuries and nonacute firearm injury narratives. They extracted the relative frequency of words and phrases (1-, 2-, 3 g) from each narrative to create the bag-of-words representation for each narrative.

We used Latent Dirichlet Allocation (LDA) to create a set of 50 topics from the narratives based on prior studies.¹⁹

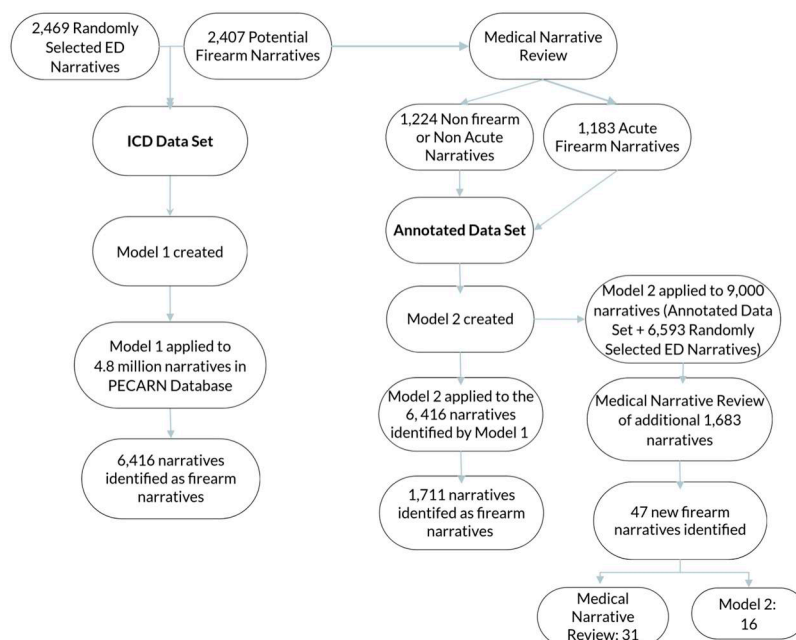


FIGURE 1. Flow diagram of medical narrative review and model creation.

LDA uses an unsupervised clustering algorithm to identify latent topics in large quantities of text.²⁰ The algorithm assumes that each word occurrence can be attributed to 1 or more topics generated from the corpus. Topics are groups of words clustered based on their co-occurrence across the corpus. The number of topics is set a priori; for this study, we generated 50 data-driven topics from all narratives in the ICD data set using LDA. The distribution of LDA topics was extracted for both data sets; topics were reviewed by the research team for internal validity.

Identifying Statistically Significant Words and Phrases Associated With Firearm Injuries

We assigned a binary category to the narratives depending on whether they were an acute firearm injury or not (0 for nonfirearm, 1 for firearm). Logistic regression was used to identify topics significantly associated with the different categories of narratives (eg, firearm vs. not). The effect size was measured using odds ratios with 95% CIs. We used Benjamini-Hochberg *P*-correction and *P* < 0.05 to indicate meaningful associations. This process was repeated for both the ICD data set and the annotated data set to understand the distribution of false positives.

Predictive Modeling

To evaluate how our features predict whether a narrative is distinguished as an acute firearm injury, we tested out a combination of words and phrases (N-grams) and all LDA topics to identify the best-performing machine learning model among Logistic Regression, Random Forest, and Extra Trees Classifier. We analyzed their performance using AUC, precision, recall, and F1 score.

We evaluated both models on a testing data set of 4,849,045 narratives collected between 2012 and 2020 in the complete PECARN Registry database of ED visits. We first applied the model trained on the ICD data set (model 1 in Table 1), using a probability threshold of 0.8 to identify narratives as firearm injuries. Subsequently, we applied the model trained on the annotated data set (model 2 in Table 1) to only those narratives flagged as firearm injuries by model 1, again using a threshold of 0.8. This threshold was selected because it corresponded to the optimal point on the ROC curve, offering the best balance between true-positive and false-positive rates.

Model 2 was then applied to a random selection of narratives from the PECARN Registry database plus the annotated data set, again using a threshold of 0.8 to distinguish acute firearm injury narratives. Then, members of the research team reviewed all narratives newly identified as acute firearm injuries by NLP model 2 and a random selection of the charts identified as nonfirearm narratives. Team members used the same criteria as the initial medical narrative review. This was performed to assess if there were narratives for acute firearm injury in the random selection of PECARN Registry narratives that had not been

identified by ICD code previously. Sensitivity and specificity of ICD codes and NLP for the identification of acute firearm injury compared with medical narrative review were then calculated.

RESULTS

A total of 2407 ED visits for potential firearm injury were identified in the PECARN Registry database using the ICD codes for potential firearm injuries. An additional randomly selected 2469 narratives without an associated ICD code for a potential firearm injury were included, creating the ICD data set (*n* = 4876). Of the 2407 narratives, 1183 narratives were annotated by the medical team review as acute firearm injuries, and 1224 were nonfirearm or nonacute injuries, creating the annotated data set.

Table 1 shows the characteristics of the 2 trained Random Forest models: 1 trained on the ICD data set and the other on the annotated data set. We found that Random Forest performed the best across a 10-fold cross-validation setting with a combination of 1-grams and Topics, with both the ICD-trained and annotated-data-trained models achieving AUC ROC > 0.977. We also tried Bidirectional Encoder Representations from Transformers (BERT) embeddings and Linguistic Inquiry and Word Count (LIWC) as additional features to predict, but did not find significant gains compared with the much higher computational needs in computing the features.^{21–23} The results of 1-grams, topics, BERT embeddings, and LIWC as features for both Random Forest models can be found in Supplemental Table 2, Supplemental Digital Content 1, <https://links.lww.com/PEC/B698>.

Applying the Random Forest model trained on the ICD data set (model 1) to the full testing data set yielded an AUC of 0.969, with 6416 of the ~4.8 million narratives classified as firearm injuries (including both true positives and false positives) (Table 2). When the second Random Forest model, trained on the annotated data set (model 2), was applied to these 6416 narratives, it identified 1711 narratives as firearm injuries, achieving an AUC of 0.664. Refer to Table 2 for the complete results of both model evaluations. Model 2 was then applied to a data set consisting of 9000 medical narratives: 2407 from the annotated data set and 6593 randomly selected from the PECARN Registry database, giving us our final NLP predictions, again based on a 0.8 cutoff.

Figure 2 shows the top themes and topics identified by NLP in potential firearm narratives in the ICD data set. Words such as “gunshot,” “wound,” and “gsw” show a strong association with medical narratives indicating potential firearm injuries, as identified by the ICD codes. However, some top topics, like “bb,” are false positives, appearing in narratives about BB guns classified as potential firearm injuries by ICD codes.

Figure 3 shows the top themes and topics identified by NLP for suspected acute firearm narratives in the

TABLE 1. Prediction Performance of Random Forest Model on the ICD Data Set and the Annotated Data Set at Classifying Firearm Injury Related Narratives With 1-Grams and Topics as Features

Random Forest (RF) Classifier With 1-Grams+Topics	AUC ROC	Precision	Recall	F1 Score
Model 1: RF on ICD data set	0.995	0.972	0.972	0.972
Model 2: RF on annotated data set	0.977	0.922	0.922	0.922

AUC ROC indicate the area under the receiver operating characteristic curve.

TABLE 2. Performance Evaluation of Random Forest Models Trained on the ICD and Annotated Data Sets on the Test Data Set

Model Evaluation	AUC ROC	Precision	Recall	F1 Score
Evaluation 1: model 1 applied to full test data set (~4.8M narratives)	0.969	0.363	0.939	0.523
Evaluation 2: model 2 applied to model 1-positive narratives (6416 narratives)	0.664	0.647	0.476	0.548

AUC ROC indicate the area under the receiver operating characteristic curve.

annotated data set. We see similar words as in Figure 2, such as “gunshot,” “wound,” and “gsw,” that are strongly associated with medical narratives with acute firearm injuries. We also see an elimination of the false-positive words, such as “bb,” which were seen in analysis with the ICD data set. These false-positive words and topics, such as “bb,” “ophthalmology,” and “fever,” are now strongly linked to nonfirearm-related narratives in this annotated data set, compared with those that had an ICD code related to firearm injury (Fig. 4).

To determine the accuracy of the approaches, the random narratives on which model 2 was applied then underwent medical narrative review for sensitivity and specificity calculations (n=3957). This included all narratives marked as an acute firearm injury by model 2 without a corresponding firearm-associated ICD code and a selection marked by model 2 as a nonacute firearm injury. NLP newly identified 16 narratives describing an acute firearm injury; 11 of those had a firearm-associated ICD code but not an acute firearm injury ICD code. An additional 31 acute firearm injuries were identified in the medical narrative review of charts marked by NLP as a nonacute firearm

injury and also lacking an associated firearm injury ICD code. This is a total of 1230 acute firearm injuries confirmed by medical narrative review. As there were a few additional acute firearm injuries discovered after reviewing 1683 narratives, the medical narrative review was discontinued.

ICD codes correctly identified 1171 (95.2%) of the 1230 acute firearm injuries in the final data set. It had a sensitivity of 95.2% (93.8% to 96.3%) and a specificity of 93.5% (92.6% to 94.4%). Of the false positives (176), the most common categories were 82 (46%) subacute injuries or complications of a firearm injury, 36 (20%) nonpowder firearms, 13 (7%) pistol whippings, 15 (8%) toys (Nerf guns, water guns, sling shots etc.) and 4 (2%) less lethal police weapons (rubber bullets, tasers). Seventy-three charts had conflicting ICD codes (either for both acute and subsequent or both for firearm and nonpowder firearm injuries) and were marked as not an acute firearm injury. Of those, 10 were acute firearm injuries.

NLP correctly identified 1041 (84.6%) of the 1230 acute firearm injuries in the final data set. It had a sensitivity of 84.6 (82.5% to 86.6%) and a specificity of 94.7% (93.8% to 95.5%). Of the false positives (146), the most

**FIGURE 2.** Words and phrases (A) and topics (B) are significantly associated with firearm narratives in the ICD data set.

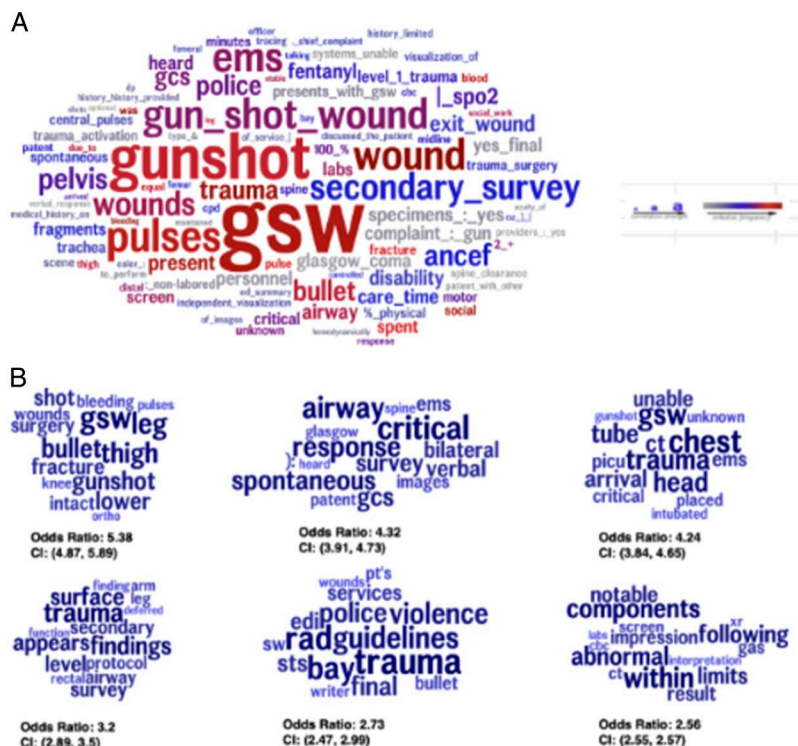


FIGURE 3. Words and phrases (A) and themes (B) are significantly associated with acute firearm injuries in the annotated data set.

common categories were other 132 (91%), traumatic injuries (motor vehicle collisions, pedestrian struck, etc.), 11 (8%) other violent injuries (stabbing, nonaccidental trauma), and 2 (1%) subacute firearm injuries.

If both NLP and ICD indicate an acute firearm injury, this correctly identifies 1040 (84.5%) of the acute firearm injuries and correctly eliminates 2401 nonacute firearm medical narratives. Three narratives are incorrectly identified by both ICD code and NLP as an acute firearm injury (false positive), 31 acute firearm narratives are missed by both NLP and ICD code (false negatives), and 483 narratives ICD and NLP disagree (163 false negatives and 320 true negatives). This method has a sensitivity of 84.5% (82.2% to 86.3%) and a specificity of 99.8% (99.6% to 100%). If either NLP or ICD code indicates an acute firearm injury, it correctly identifies 1202 of the acute firearm injuries (true positives) and correctly eliminates 2401 encounters (true negatives). There are 323 false positives and 31 false negatives. This method has a sensitivity of 97.5% (96.4% to 98.3%) and a specificity of 88.1% (86.9% to 89.3%).

DISCUSSION

Both ICD codes and NLP models had strong sensitivity and specificity for identifying acute firearm injuries in this multicenter data set. ICD code identification had difficulty distinguishing subacute firearm injuries or nonpowder firearm injuries from firearm injuries. It is also notable that some medical narratives ($n = 73$) had contradictory ICD codes associated with a visit. Researchers should be aware of this issue as it could affect the specificity

of ICD code identification and adjust their methods accordingly.

False positives while using NLP were mostly other traumatic injuries. This likely occurred because multiple topics associated with acute firearm injury are also associated with other traumatic injuries, such as “critical” or “injury” (Fig. 3). Given that there is 1 main subset of narratives that are misidentified by NLP, future models could be modified to identify and exclude more traumatic injuries, similar to the development of model 2 to eliminate nonpowder firearm injuries and subacute injuries. If improved, NLP could become more useful as a primary method of identifying acute firearm injuries.

It is potentially useful that ICD and NLP are complementary methods of identifying acute firearm injuries. When using both methods to identify acute firearm injuries, only 31 acute firearm injuries of the 1230 total went unidentified by both methods. Using both NLP and ICD codes reduced a data set of almost 4000 charts to only 513 that needed to be manually reviewed. This reduction in expensive and time-consuming work could be useful if researchers had access to the medical narrative and wanted to ensure they had the most complete capture of acute firearm injuries in their data set.

However, these findings also suggest that the ICD codes in isolation have reasonable sensitivity and specificity for identifying acute firearm injuries in a pediatric population. This finding should be reassuring to the scientific community that has relied on ICD codes to identify acute firearm injuries for research purposes, especially in large data sets, which often do not provide access to narrative data. NLP models require access to medical narratives,

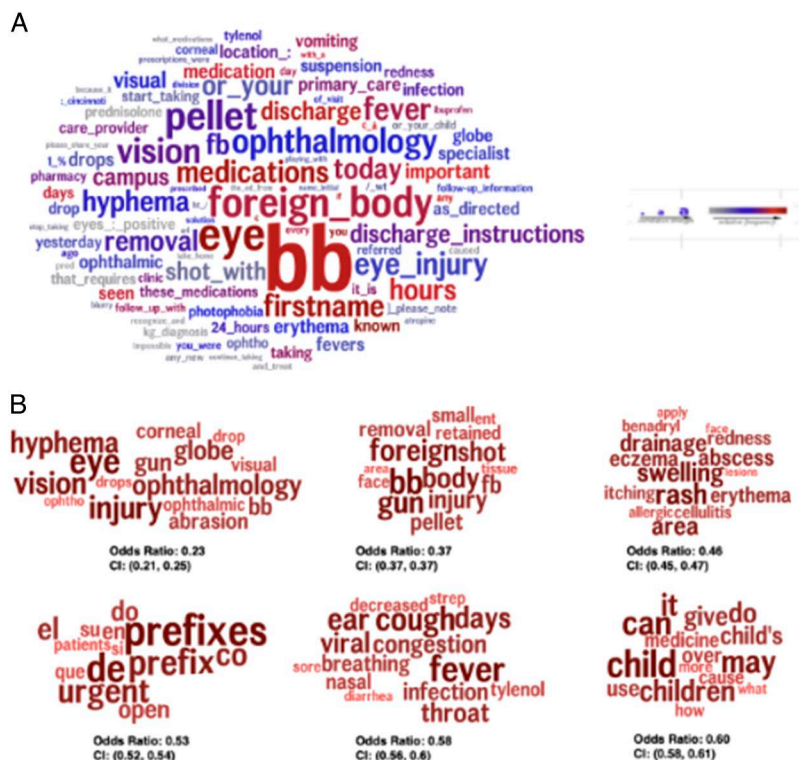


FIGURE 4. Words and phrases (A) and topics (B) are significantly associated with nonfirearm narratives in the annotated data set.

specialized experience, and data processing capabilities that may be out of reach for many researchers. This work suggests that for the simple identification of firearm injury, the ICD code is sufficient.

It is important to point out that for the interpretation of our findings, the gap between the high cross-validation performance of the NLP models (AUC > 0.977) and the lower AUC of the second-stage NLP model (0.664) reflects the cascaded design rather than overfitting. Because model 2 was applied only to narratives already flagged by model 1, it was evaluated on a population from which straightforward nonfirearm narratives had been removed, leaving predominantly confusable cases (subacute injuries, non-powder firearms, and other trauma). Discriminating true acute firearm injuries within this residual set is inherently harder than the original balanced classification, and the first-stage model's AUC of 0.969 on the full registry indicates the pipeline generalized well overall.

This study does have limitations that deserve discussion. One, sensitivity and specificity were calculated on an “enhanced sample” of medical narratives with an increased incidence of firearm injuries. Because sensitivity and specificity are conditional on true case status, they are not biased by this enrichment; however, at the much lower prevalence of firearm injury in routine clinical data, the same specificity may yield more false positives per true case, lowering positive predictive value in real-world deployment. Also, this data was derived from 7 tertiary-care pediatric EDs. This patient population likely overselects for severe firearm injury, which may affect the results. Providers at tertiary-care pediatric EDs may use language in

their medical narratives that is not used in general EDs, which could affect the generalizability of these results. This analysis was also limited by the number of medical narratives the research team could review. Although shared training and a standardized template were used, we did not calculate interrater reliability in narrative reviews. It is possible that more firearm injuries were missed by both ICD and NLP.

Future directions should include validating this NLP algorithm in other data sets to compare sensitivity and specificity in different populations. Newer NLP methods, including foundational large language models, may perform better than the traditional machine learning methods used here, particularly for reducing the confuser-driven false positives that limited our second-stage model, 91% of which were other traumatic injuries sharing vocabulary with firearm trauma. Recent studies have also evaluated using NLP to assess the intent of firearm injury.²⁴ Given these findings and the known inaccuracies in pediatric firearm injury intent documentation with ICD codes, it is important to understand this phenomenon in a pediatric population.^{7,8} Recent work has also been done to assess using NLP to identify firearm exposure (hearing shots, knowing someone who has been shot) from medical record narratives.²⁵ This method should be applied to a pediatric population, who may be even more affected by exposure to firearm violence, but this stressor often goes unaddressed in the pediatric ED.²⁶

In this comparison of NLP models derived from medical record narratives and ICD codes to identify acute firearm injury in this multicenter data set, both models

correctly identified acute firearm injury with strong sensitivity and specificity. These findings support further use of ICD for the identification of acute firearm injury and add to a growing body of literature evaluating the value of NLP in expanding pediatric firearm injury research.

PECARN Registry Study Group and the FACTS Consortium

Members of the PECARN Registry Study Group include Drs Elizabeth Alpern (Lurie Children's Hospital), Lynn Babcock (Cincinnati Children's Hospital), Lalit Bajaj (University of Colorado/Children's Hospital Colorado), David Brousseau (Nemours Children's Health/Sidney Kimmel Medical College at Thomas Jefferson University), James Chamberlain (Children's National Hospital), Robert Grundmeier (Children's Hospital of Philadelphia), Bashar Shihabuddin (Nationwide Children's Hospital), and Joseph J. Zorc (Children's Hospital of Philadelphia). Members of the FACTS Consortium include Drs Rebecca Cunningham (University of Minnesota), Patrick Carter (University of Michigan), Marc Zimmerman (University of Michigan) and Lynn Massey (University of Michigan).

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